

Conditional Exponential DLPP with Flat Initial Condition

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$$E(\pi) := \sum_{(i,j) \in \pi} w_{i,j}, \quad \mathcal{L}^{\text{ptp}}(m,n) := \max_{\substack{\pi: (0,0) \rightarrow (m,n) \\ \pi \text{ up-right}}} E(\pi).$$

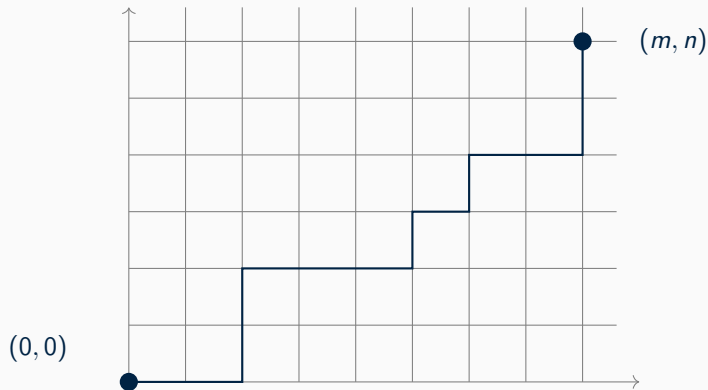


Figure 1: Exponential Directed Last Passage Percolation (DLPP) [Bai22]

Law of Large Numbers for Point-to-point DLPP [Ros81]:

$$\lim_{N \rightarrow \infty} \frac{\mathcal{L}^{ptp}(xN, yN)}{N} = (\sqrt{x} + \sqrt{y})^2$$

[BCT25] conditioned distribution upon the event that a given point (aN, bN) satisfies

$$\mathcal{L}^{ptp}(aN, bN) = \ell N > (\sqrt{a} + \sqrt{b})^2 N.$$

[BCT25] also considered fluctuations for point-to-point DLPP, which we will do later.

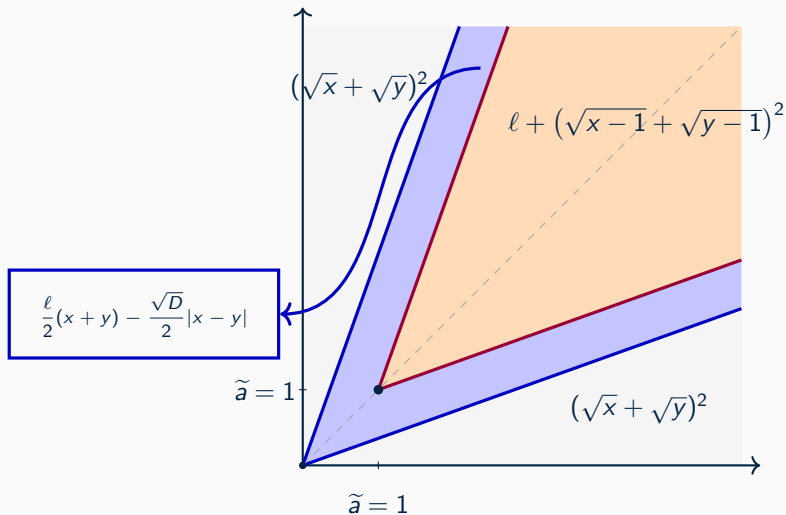


Figure 2: Three Law of Large Number Regimes in Step Case

$$\mathcal{L}(m, n) := \max_{\pi: (-k, k) \rightarrow (m, n)} E(\pi).$$

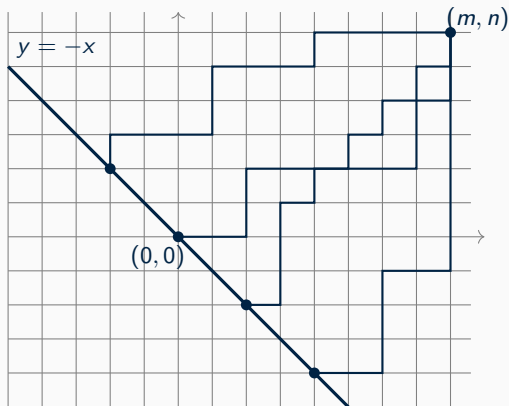


Figure 3: Flat Exponential DLPP

Part 1: Conditional Law of Large Numbers Heuristic

Part 2: Rigorous Computation of Fluctuations along Diagonal

Part 3: Heuristic for General Fluctuations

Conditional Law of Large Numbers Heuristic

Conditioning on Upper Large Deviation Event

Law of Large Numbers for Line-to-point DLPP [Joh99]:

$$\lim_{N \rightarrow \infty} \frac{\mathcal{L}(xN, yN)}{N} = 2(x + y)$$

Now we condition our distribution upon the event that a given point (aN, bN) satisfies

$$\mathcal{L}(aN, bN) = \ell N > 2(a + b)N.$$

We now follow a similar heuristic argument as in [BCT25].

Conditional Law of Large Numbers Heuristic

$$H(t) = t\ell + \left(\sqrt{(x-t)\tilde{a}} + \sqrt{(y-t)\tilde{a}} \right)^2$$
$$\lim_{N \rightarrow \infty} \frac{\mathcal{L}(x\tilde{a}N, y\tilde{a}N)}{N} = \max\{2(x\tilde{a} + y\tilde{a}), H(t) : 0 \leq t \leq \min(x, y, 1)\}$$

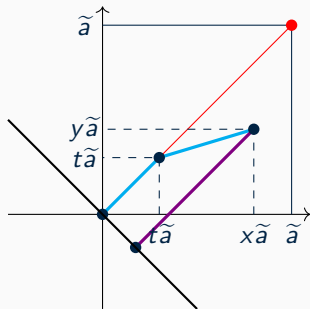


Figure 4: Conjectural maximizing path

Conditional Law of Large Numbers Heuristic (cont.)

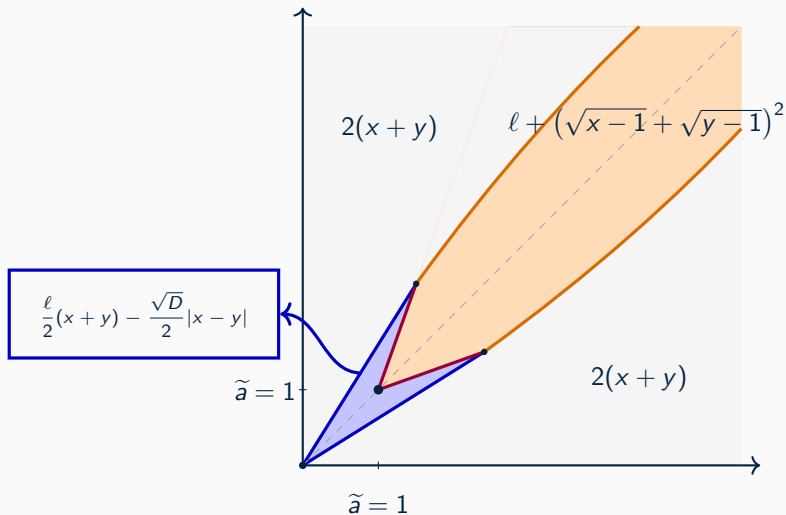


Figure 5: Three Law of Large Number Regimes in Flat Case

Compare with Point-to-Point Regimes

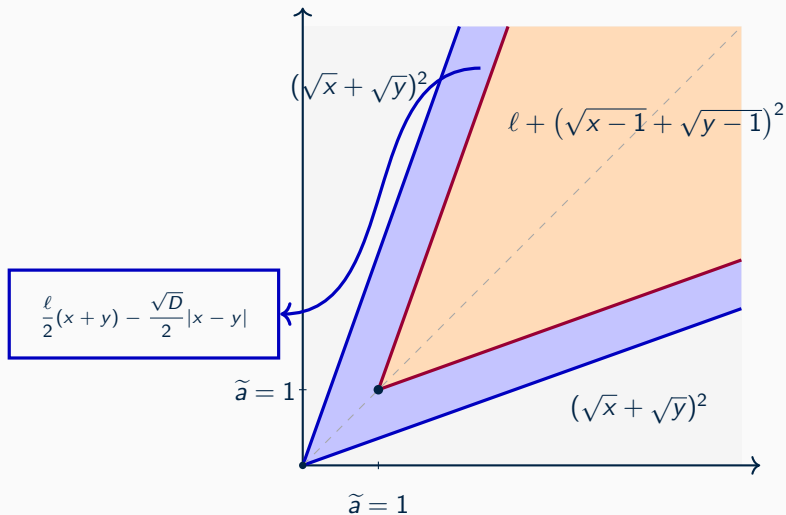
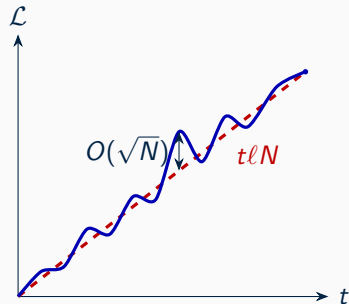


Figure 6: Three Law of Large Number Regimes in Step Case

Rigorous Computation of Fluctuations Along Diagonal

Fluctuations as a process-level central limit theorem

Limit shape and random fluctuation



Dashed: the deterministic $O(N)$ profile.

Blue: an $O(\sqrt{N})$ random deviation.

Comparison with the usual CLT

$$\frac{\sum_{i=1}^N X_i - N\mathbb{E}[X_i]}{\sigma\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1)$$

$$\frac{\mathcal{L}(\text{near}(taN, taN)) - t\ell N}{\sigma\sqrt{N}} \xrightarrow{d} X_N(t, s)$$

We have a formula for the conditional distribution from Proposition 2.3 in [Liu21]. Writing this with notation as in [BCT25], we have

$$\mathbb{P}(\mathcal{L}(M_1, N_1) > T_1 | \mathcal{L}(M_2, N_2) = T_2) = \frac{Q_2^{flat}(\mathbf{M}, \mathbf{N}, \mathbf{T})}{Q_1^{flat}(M_2, N_2, T_2)},$$

where

$$Q_2^{flat}(\mathbf{M}, \mathbf{N}, \mathbf{T}) = \sum_{\mathbf{n} \in \mathbb{N}^2} \frac{1}{(\mathbf{n}!)^2} Q_2^{(\mathbf{n}), flat}(\mathbf{M}, \mathbf{N}, \mathbf{T}),$$

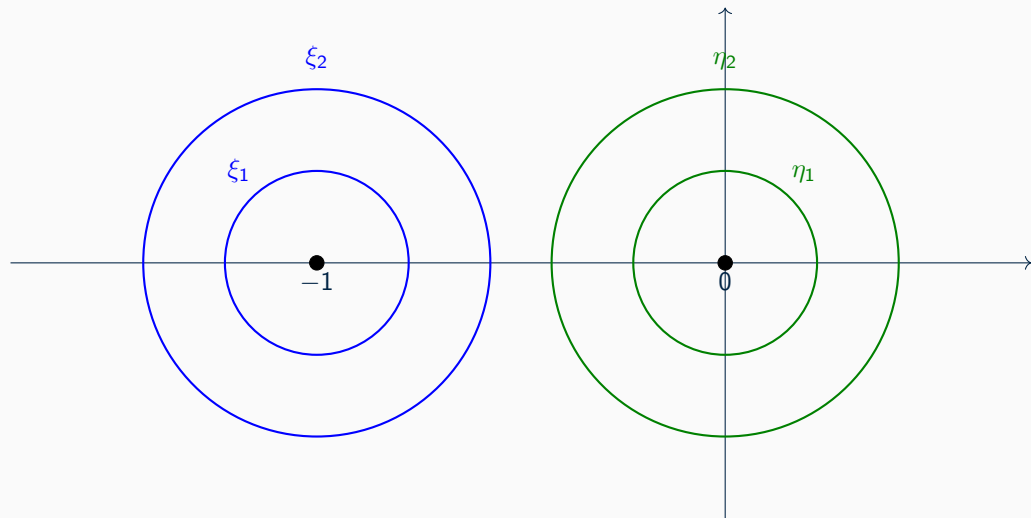
and

$$Q_2^{(1,1), flat}(\mathbf{M}, \mathbf{N}, \mathbf{T}) = \frac{1}{2\pi i} \oint_{>1} \tilde{D}_{\mathbf{M}, \mathbf{N}, \mathbf{T}}^{(1,1), flat}(z) \frac{1}{(z+1)z^2} dz.$$

$$\tilde{D}_{\mathbf{M},\mathbf{N},\mathbf{T}}^{(1,1),flat}(z)$$

$$\tilde{D}_{\mathbf{M},\mathbf{N},\mathbf{T}}^{(1,1),flat}(z) = \frac{-1}{(2\pi i)^4} \left[\int_{C_{2,L}^{in}} d\xi_2 + z \int_{C_{2,L}^{out}} d\xi_2 \right] \left[\int_{C_{2,R}^{in}} d\eta_2 + z \int_{C_{2,R}^{out}} d\eta_2 \right] \\ \left[\int_{C_{1,L}} d\xi_1 \right] \left[\int_{C_{1,R}} d\eta_1 \right] \tilde{\Pi}_{(1,1),flat}(\xi, \eta) F_{\mathbf{M},\mathbf{N},\mathbf{T}}^{(1,1)}(\xi, \eta) \delta(-\eta_1 - 1, \xi_1)$$

Contours

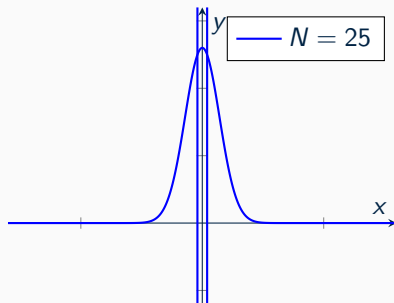
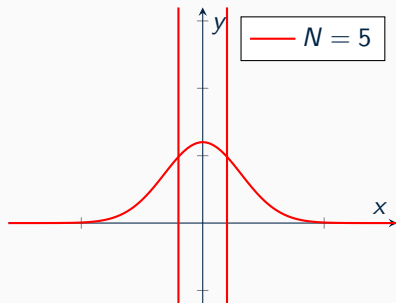


$$\tilde{D}_{\mathbf{M},\mathbf{N},\mathbf{T}}^{(1,1),flat}(z)$$

$$\tilde{D}_{\mathbf{M},\mathbf{N},\mathbf{T}}^{(1,1),flat}(z) = \frac{-1}{(2\pi i)^4} \left[\int_{C_{2,L}^{in}} d\xi_2 + z \int_{C_{2,L}^{out}} d\xi_2 \right] \left[\int_{C_{2,R}^{in}} d\eta_2 + z \int_{C_{2,R}^{out}} d\eta_2 \right] \\ \left[\int_{C_{1,L}} d\xi_1 \right] \left[\int_{C_{1,R}} d\eta_1 \right] \underbrace{\tilde{\Pi}_{(1,1),flat}(\xi, \eta)}_{\text{Rational Function}} \underbrace{F_{\mathbf{M},\mathbf{N},\mathbf{T}}^{(1,1)}(\xi, \eta)}_{\text{Exponentials}} \underbrace{\delta(-\eta_1 - 1, \xi_1)}_{\text{Dirac } \delta\text{-kernel}}$$

Laplace's Method

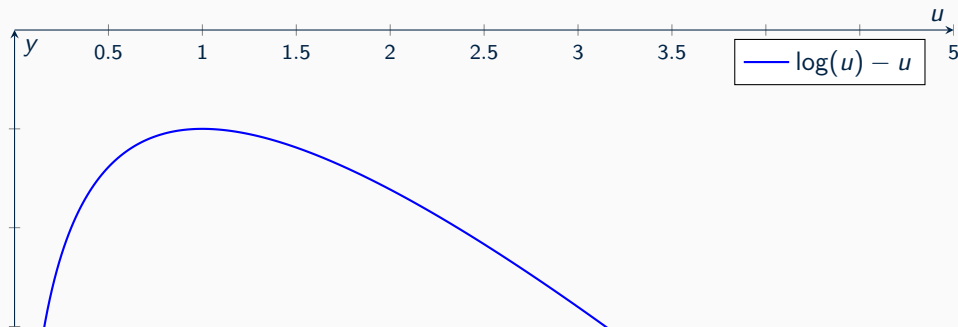
$$\lim_{N \rightarrow \infty} \left(\int_{-\delta_N}^{\delta_N} e^{N(G(x_{\text{crit}}) + \frac{1}{2}G''(x_{\text{crit}})x^2 + O(x^3))} dx + \int_{\mathbb{R} \setminus (-\delta_N, \delta_N)} e^{NG(x)} dx \right)$$



Example: Stirling's Formula

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt = x^{x+1} \int_0^\infty e^{x(\log u - u)} du$$

and when $x \in \mathbb{N}$, $\Gamma(x+1) = x!$



$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

Steepest Descent: Local Picture Near a Critical Point z_c

Want to apply Laplace's method in complex plane.

$$\exp(G(z)) = \exp(\operatorname{Re}(G(z)) + i \operatorname{Im}(G(z))) = \exp(\operatorname{Re}(G(z))) \exp(i \operatorname{Im}(G(z)))$$

We want a contour such that:

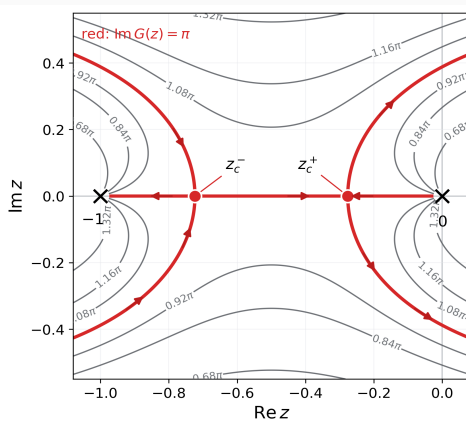
1. Imaginary part is constant
2. Real part attains local extrema

Then, Cauchy-Riemann $\implies$ we want a particular contour through critical point of $G(z)$.

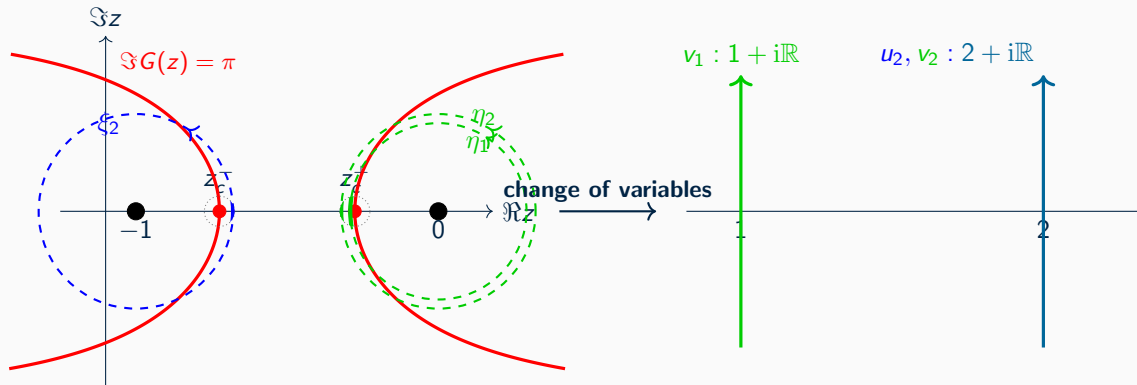
The Modified Contours

Let $G(z) = -a \log(z+1) + a \log(z) + \ell z$,

$$F_{M,N,T}^{(1,1)}(\xi, \eta) = \exp \left\{ N \left[(1-t)G(\xi_2) - (1-t)G(\eta_2) + tG(-\eta_1 - 1) - tG(\eta_1) \right] + O(N^{1/2}) \right\}$$



Analysis near the Critical Points

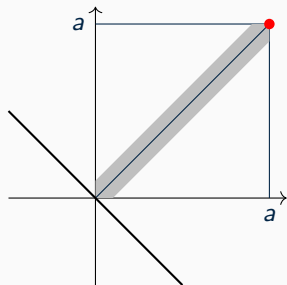


Fluctuations near Diagonal

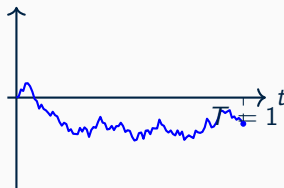
Theorem

Fix $\ell > \tilde{\mathcal{L}}(a, a) = 4a$. Then, as $N \rightarrow \infty$,

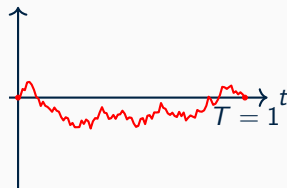
$$\text{Law} \left(\left\{ \frac{\mathcal{L}(taN + sCN^{1/2}, taN - sCN^{1/2}) - t\ell N}{\sigma N^{1/2}} \right\} \mid \mathcal{L}(aN, aN) = \ell N \right) \xrightarrow{\text{f.d.d.}} \text{Law} (\mathbb{B}_1^{\text{br}}(t) - |\mathbb{B}_2^{\text{br}}(t) - (1-t)Z - s|)$$



Standard Brownian Motion



Brownian Bridge



Fluctuations near Diagonal (cont.)

Theorem ([BCT25])

¹ Fix $\ell > \bar{\mathcal{L}}^{\text{ptp}}(a, b) = (\sqrt{a} + \sqrt{b})^2$. Then, as $N \rightarrow \infty$,

$$\text{Law} \left(\left\{ \frac{\mathcal{L}^{\text{ptp}}(taN + sCN^{1/2}, tbN - sCN^{1/2}) - t\ell N}{\sigma N^{1/2}} \right\} \middle| \mathcal{L}^{\text{ptp}}(aN, bN) = \ell N \right) \xrightarrow{f.d.d.} \text{Law} \left(\mathbb{B}_1^{\text{br}}(t) - |\mathbb{B}_2^{\text{br}}(t) - s| \right).$$

Theorem

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¹The coefficients C 's in [BCT25]'s Theorem are marked the same for simplicity. When $a = b$, all the C 's are the same.

Heuristic for General Fluctuations

Heuristic for General Fluctuations

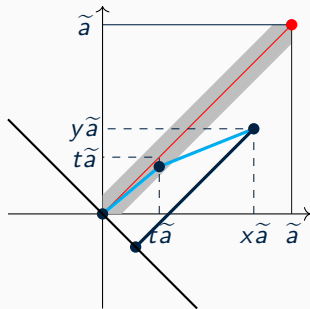


Figure 8: Conjectural general fluctuations.

Heuristic for General Fluctuations

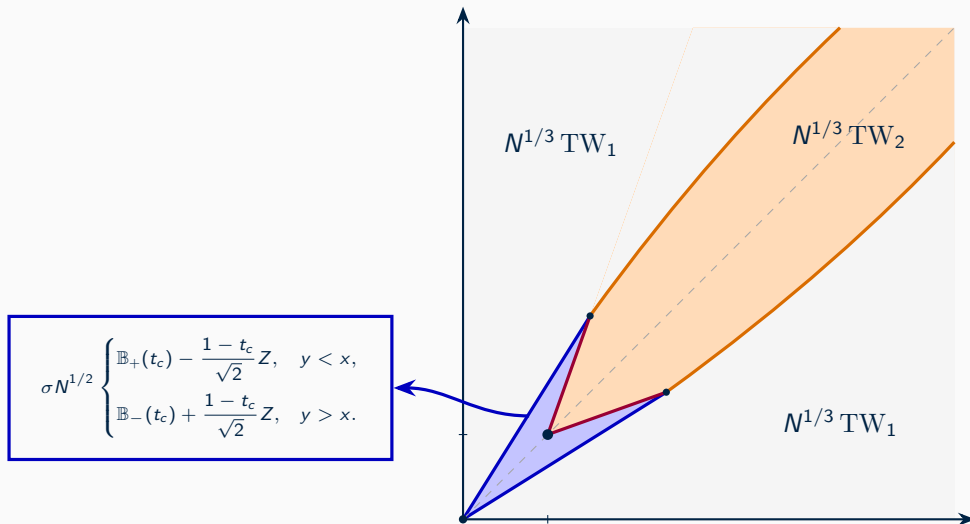
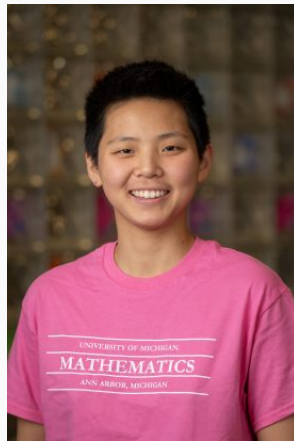


Figure 9: Three fluctuation regimes.

Acknowledgments



Prof. Baik



Tian Dong

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- [BCT25] Jinho Baik, Dylan Cordaro, and Tejaswi Tripathi, *Conditional exponential directed last passage percolation under a one-point upper large deviation event*, 2025, Version Number: 2.
- [Joh99] Kurt Johansson, *Shape Fluctuations and Random Matrices*, October 1999, arXiv:math/9903134.
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- [Ros81] H. Rost, *Non-equilibrium behaviour of a many particle process: Density profile and local equilibria*, Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete **58** (1981), no. 1, 41–53 (en).